# R86 | Frozen shared-readout geometry; diagnostic only

| Split | Rule | Top1 % | Regret % | mean_cost | Target recall % | Corrected / harmed |
|---|---|---|---|---|---|---|
| TRAIN | original | 70.8333 | 0.366116 | 11.5372107 | 0.0000 | 0 / 0 |
| TRAIN | angle-only | 71.0069 | 0.365688 | 11.5371388 | 0.0000 | 8 / 6 |
| VAL | original | 71.2674 | 0.415504 | 11.3647208 | 0.0000 | 0 / 0 |
| VAL | angle-only | 71.3542 | 0.402227 | 11.3631038 | 0.0000 | 6 / 5 |

Frozen R84A shared final Linear: there are no method-specific classifier weight norms. Only hidden-feature radius and angle to the same w are measured.

CPU FP32, 64 TRAIN + 64 VAL samples/task (or the recorded input/timing-only budget reduction), seed2. Not a formal AMP evaluation, fitted model, threshold search, TEST or LIB result.

Maximin utilities permit common shifts. Angle-only argmax is a fixed-parameter counterfactual, not a calibrated/Bayes-equivalent inference rule.

A favorable change is only priority evidence for a new end-to-end normalized-head experiment, not causal proof or an accepted inference change. A negative result does not rule out normalized architectures.

No native winners in a sampled method group means undefined recall, never zero. All18 macros equally weight tasks; mixed raw mean_cost is descriptive only.

# R86 | Sampled frozen-target method geometry

| Split / task | Method | Winners | TP orig / angle | Feature norm on wins | Cosine on wins | Rescued / new FP |
|---|---|---|---|---|---|---|
| TRAIN / TSP | LEHD | 14 | 0 / 0 | 2.7216 | -0.0720 | 0 / 0 |
| TRAIN / OVRPLTW | MVMOE | 9 | 0 / 0 | 1.9532 | -0.1239 | 0 / 0 |
| TRAIN / TSP | DIFUSCO | 7 | 0 / 0 | 3.6103 | -0.1566 | 0 / 0 |
| TRAIN / OVRPB | MTPOMO | 5 | 0 / 0 | 2.7094 | -0.2276 | 0 / 0 |
| TRAIN / OVRPB | MVMOE | 4 | 0 / 0 | 2.5758 | -0.1387 | 0 / 0 |
| TRAIN / VRPBL | MTPOMO | 4 | 0 / 0 | 2.7114 | -0.1770 | 0 / 0 |
| TRAIN / ATSP | MATPOENET | 3 | 0 / 0 | 2.6740 | 0.0019 | 0 / 0 |
| TRAIN / CVRP | RouteFinder | 3 | 0 / 0 | 6.4111 | -0.1534 | 0 / 0 |
| TRAIN / OVRP | reld_mt | 2 | 0 / 0 | 5.7559 | -0.3192 | 0 / 0 |
| TRAIN / OVRPBL | MTPOMO | 2 | 0 / 0 | 3.1808 | -0.1741 | 0 / 0 |
| TRAIN / OVRPBL | MVMOE | 2 | 0 / 0 | 4.5174 | -0.3830 | 0 / 0 |
| TRAIN / VRPL | MVMOE | 2 | 0 / 0 | 4.1026 | -0.3179 | 0 / 0 |
| TRAIN / VRPTW | RouteFinder | 2 | 0 / 0 | 6.8700 | -0.2721 | 0 / 0 |
| TRAIN / OVRPL | MTPOMO | 1 | 0 / 0 | 9.1215 | -0.3832 | 0 / 0 |
| TRAIN / OVRPLTW | MoSES_CaDA | 1 | 0 / 0 | 8.4458 | -0.4540 | 0 / 0 |
| TRAIN / OVRPTW | MoSES_CaDA | 1 | 0 / 0 | 3.2794 | -0.2179 | 0 / 0 |
| TRAIN / VRPL | MTPOMO | 1 | 0 / 0 | 7.0323 | -0.3732 | 0 / 0 |
| TRAIN / VRPLTW | MoSES_CaDA | 1 | 0 / 0 | 3.9978 | -0.2598 | 0 / 0 |

At most18 nonempty method/split examples shown by sampled winner support. Every method and every original predicted rank is in CSV.

Methods with no sampled native winners have undefined recall, including any zero-choice groups.

One shared w is used for every method; these are hidden-feature statistics, not per-class weight norms.

# R86 | Per-task sampled changes and fixed-representation tradeoffs

| Task | TRAIN dTop1 pp | TRAIN dRegret pp | Better angle / missed | VAL dTop1 pp | VAL dRegret pp | Better angle / missed |
|---|---|---|---|---|---|---|
| TSP | 0.00 | 0.0002 | 0 / 38 | 0.00 | 0.0000 | 0 / 36 |
| CVRP | 1.56 | -0.0449 | 2 / 36 | 4.69 | -0.2411 | 4 / 29 |
| ATSP | 1.56 | 0.0182 | 2 / 14 | 0.00 | 0.0000 | 0 / 11 |
| OVRP | 3.12 | -0.0479 | 2 / 39 | 0.00 | 0.0000 | 0 / 37 |
| VRPB | 0.00 | 0.0000 | 0 / 10 | -1.56 | 0.0569 | 0 / 15 |
| VRPL | 0.00 | 0.0334 | 1 / 41 | 0.00 | -0.0081 | 1 / 42 |
| VRPTW | 0.00 | 0.0000 | 0 / 16 | 0.00 | 0.0000 | 0 / 9 |
| OVRPTW | 0.00 | 0.0000 | 0 / 12 | 0.00 | 0.0000 | 0 / 12 |
| OVRPB | 0.00 | 0.0000 | 0 / 9 | 0.00 | 0.0000 | 0 / 10 |
| OVRPL | -3.12 | 0.0334 | 1 / 34 | -1.56 | -0.0467 | 1 / 37 |
| VRPBL | 0.00 | 0.0000 | 0 / 19 | 0.00 | 0.0000 | 0 / 18 |
| VRPBTW | 0.00 | 0.0000 | 0 / 15 | 0.00 | 0.0000 | 0 / 15 |
| VRPLTW | 0.00 | 0.0000 | 0 / 9 | 0.00 | 0.0000 | 0 / 12 |
| OVRPBL | 0.00 | 0.0000 | 0 / 4 | 0.00 | 0.0000 | 0 / 5 |
| OVRPBTW | 0.00 | 0.0000 | 0 / 10 | 0.00 | 0.0000 | 0 / 14 |
| OVRPLTW | 0.00 | 0.0000 | 0 / 10 | 0.00 | 0.0000 | 0 / 6 |
| VRPBLTW | 0.00 | 0.0000 | 0 / 13 | 0.00 | 0.0000 | 0 / 14 |
| OVRPBLTW | 0.00 | 0.0000 | 0 / 7 | 0.00 | 0.0000 | 0 / 9 |

Better angle / missed: native winner has higher cosine than the originally selected method but loses original utility. This need not make it angle-argmax.

The utility bias/common-shift freedom prevents treating this decomposition as causal or uniquely identifiable.

No TEST/LIB, fitting, threshold/coefficient search, solver execution or GPU use. The research goal remains active.